Wilhem Barbier

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Education

PhD in Computer Graphics
Université de Toulouse (2022-2026)

My research focuses on designing GPU algorithms to build and update data structures for real-time raytracing

MSc in Computer Science and Applied Math
Ensimag, Grenoble (2020-2022)

Followed the *Mathematical Models, Graphics and Simulation* track, with a specialization in Computer Graphics

BSc in Computer Science and Mathematics
Université de Toulouse (2017-2020)

Experience

Research intern, Adobe
March 2024-August 2024 (Paris, France)

Designed and implemented GPU algorithms for efficient raytracing of signed distance fields

This work was integrated within a consumer-facing product

Research intern, Unity Technologies
February 2022-August 2022 (Grenoble, France)

Worked on extending Ptex (used in texture painting workflows) to non-quad meshes in a GPU-friendly manner

R&D intern, ZEILT productions
June 2021-September 2021 (Remote)

Experimented with automated generation of lipsync animation from audio and text

Contributed to pipeline automation and solved rendering issues on an active production

Part-time research intern, INRIA Grenoble
January 2020-May 2020 (Grenoble, France)

Implemented the initial prototype for a novel real-time global illumination solver

Tech

Graphics APIs: **OpenGL 3+**, **Vulkan**
GLSL for shader development
GPGPU programming with **CUDA**
Low-level programming using **C++**
Performance analysis tools: **perf**, **NVIDIA Nsight**
Python for scripting tasks

Languages

English
Fluent

French
Mother tongue

Publications

Fused Collapsing for Wide BVH Construction
High Performance Graphics 2024

Lipschitz Pruning for Hierachical Simplification of SDFs
Eurographics 2024

HTex: Per-Halfedge Texturing for Arbitrary Mesh Topologies
High Performance Graphics 2022

A Data-Driven Paradigm for Precomputed Radiance Transfer
High Performance Graphics 2022